

GrounUp Enterprise Specification

Phase 03 - Enterprise Data Model

Version 1.0.0-review | Review-ready | 2026-07-24

Purpose

Canonical enterprise data language for every GrounUp module, AI workflow, integration, and report.

Data Domains

Foundation: Tenant, organization, legal entity, business unit, location, project, user, team, role, and reference data

Party & Relationship: Customers, prospects, vendors, subcontractors, employees, contacts, addresses, and relationship roles

Work & Delivery: Projects, opportunities, estimates, contracts, work orders, tasks, schedules, daily reports, and approvals

Resources: Labor, crews, equipment, fleet, materials, services, vendors, rates, and capabilities

Financial: Cost codes, budgets, commitments, invoices, payments, journal references, taxes, currencies, and cost transactions

Documents & Records: Files, documents, versions, attachments, classifications, retention, signatures, and evidence

Location & Spatial: Sites, parcels, coordinates, geometry references, stationing, surfaces, zones, and jurisdiction

Safety & Quality: Incidents, inspections, observations, corrective actions, permits, qualifications, and quality records

AI & Automation: Agents, prompts, models, evidence, recommendations, approvals, writebacks, and feedback

Integration & Audit: External identifiers, source systems, mappings, imports, events, lineage, changes, and immutable audit

Requirements

- EDM-000001 - Canonical model.** The platform shall maintain one governed canonical enterprise data model shared by all modules.
- EDM-000002 - Tenant key.** Every tenant-owned record shall include a verified tenant identifier.
- EDM-000003 - Global identifiers.** Every governed entity shall use immutable globally unique identifiers.
- EDM-000004 - Versioning.** Versioned entities shall preserve historical versions and effective dates.
- EDM-000005 - Soft deletion.** Business records shall use governed lifecycle states rather than uncontrolled physical deletion.
- EDM-000006 - Audit columns.** Every mutable record shall capture creation and last-change actor and timestamp.
- EDM-000007 - Optimistic concurrency.** Mutable records shall support version checks to prevent lost updates.
- EDM-000008 - Reference data.** Codes and enumerations shall be represented through governed reference sets.
- EDM-000009 - Custom fields.** Tenant custom fields shall use typed governed definitions and values.
- EDM-000010 - Data ownership.** Each entity and field shall have an accountable data owner.
- EDM-000011 - System of record.** Each governed field shall declare its authoritative source.
- EDM-000012 - Null semantics.** Required, optional, unknown, not applicable, and withheld values shall be distinguishable where material.
- EDM-000013 - Precision.** Numeric values shall store declared precision, scale, and unit or currency context.
- EDM-000014 - Time handling.** Timestamps shall use UTC with source timezone where business meaning requires it.
- EDM-000015 - Effective dating.** Rates, assignments, relationships, and classifications shall support effective-date ranges.
- EDM-000016 - Natural keys.** External or business keys shall not replace immutable internal identifiers.
- EDM-000017 - Cross-entity references.** References shall enforce compatible tenant and lifecycle constraints.
- EDM-000018 - Record classification.** Every entity shall inherit or define data security classification.
- EDM-000019 - Retention.** Records shall map to retention and legal-hold policies.
- EDM-000020 - Data lineage.** Imported, calculated, converted, and AI-generated values shall preserve lineage.
- EDM-000021 - Source identity.** Imported records shall preserve source system and external identifier.
- EDM-000022 - Duplicate prevention.** Party, vendor, customer, project, and resource master data shall support duplicate detection.
- EDM-000023 - Merge controls.** Master-record merges shall preserve aliases, lineage, audit, and reversal evidence.
- EDM-000024 - Relationship model.** Parties and business objects shall support typed effective-dated relationships.
- EDM-000025 - Organization hierarchy.** Legal entities, business units, branches, teams, and projects shall support hierarchical relationships.
- EDM-000026 - Project hierarchy.** Projects shall support programs, parent projects, phases, areas, and work packages.
- EDM-000027 - Work breakdown.** Work records shall support WBS and cost-code structures without hardcoding one industry.
- EDM-000028 - Estimate immutability.** Released estimate versions shall be immutable.
- EDM-000029 - Estimate traceability.** Estimate lines shall retain quantity, production, resource, rate, markup, and source provenance.
- EDM-000030 - Resource polymorphism.** Resource assignments and rates shall support labor, crew, equipment, material, and service resource types.
- EDM-000031 - Rate effective dates.** Rates shall include location, source, currency, unit, and effective periods.
- EDM-000032 - Financial separation.** Budget, commitment, actual, forecast, invoice, and payment records shall remain distinct.
- EDM-000033 - Cost transaction integrity.** Cost transactions shall balance required dimensions and preserve source documents.
- EDM-000034 - Currency integrity.** Money values shall always include currency.
- EDM-000035 - Unit integrity.** Quantities shall always include unit and dimension.
- EDM-000036 - Tax jurisdiction.** Tax calculations shall reference explicit jurisdiction and rule versions.
- EDM-000037 - Document logical identity.** Documents shall have a stable logical identity independent of content versions.
- EDM-000038 - Document version immutability.** Accepted document versions shall be immutable and checksummed.
- EDM-000039 - Attachment model.** Documents shall link to any supported business object without duplicating content.
- EDM-000040 - Signature evidence.** Electronic signatures shall preserve signer, method, timestamp, hash, and document version.
- EDM-000041 - Spatial reference.** Geometry shall declare its coordinate reference system.
- EDM-000042 - Spatial abstraction.** Business entities shall reference geometry without forcing all geometry into transactional tables.
- EDM-000043 - Stationing.** Linear projects shall support station ranges and offsets.
- EDM-000044 - Jurisdiction mapping.** Projects, sites, taxes, permits, and compliance records shall reference jurisdiction.
- EDM-000045 - Safety linkage.** Incidents, inspections, observations, and corrective actions shall link to project, location, people, and evidence.
- EDM-000046 - Qualification history.** Qualifications shall preserve issuer, type, dates, status, and evidence.

EDM-000047 - AI provenance. AI-created or modified values shall identify agent, model, prompt, evidence, and execution.

EDM-000048 - AI writeback separation. AI recommendations shall remain separate from committed business records until authorized writeback.

EDM-000049 - Feedback linkage. Feedback shall link to the exact system or AI output version being evaluated.

EDM-000050 - Event envelope. Domain events shall use a standard tenant-aware versioned envelope.

EDM-000051 - Change data capture. Material record changes shall emit change metadata or domain events.

EDM-000052 - API contract alignment. API schemas shall map directly to canonical entities or declared view models.

EDM-000053 - Reporting model. Operational entities shall support governed reporting views without duplicating source truth.

EDM-000054 - Search model. Search documents shall remain derivative and traceable to canonical records.

EDM-000055 - Import staging. Imports shall validate in staging before committed writes.

EDM-000056 - Export completeness. Exports shall include identifiers, versions, units, currencies, and lineage needed for interpretation.

EDM-000057 - Schema evolution. Breaking schema changes shall require versioned migration and compatibility plans.

EDM-000058 - Backward compatibility. Published event and API contracts shall follow declared compatibility windows.

EDM-000059 - Data quality rules. Critical fields shall have completeness, validity, uniqueness, consistency, and timeliness measures.

EDM-000060 - Data quality exceptions. Failed quality rules shall create reviewable exceptions rather than silent correction.

EDM-000061 - Masking. Sensitive fields shall support masking and tokenization policies.

EDM-000062 - Encryption attributes. Highly sensitive data shall support field-level encryption metadata where required.

EDM-000063 - Regional residency. Entity storage classes shall support region and residency constraints.

EDM-000064 - Archival tiers. Archived records shall retain resolvable identifiers and access controls.

EDM-000065 - Legal hold. Legal hold shall suspend deletion and archival expiration.

EDM-000066 - Referential integrity. Deletion or archival shall not orphan governed references.

EDM-000067 - Materialized views. Materialized reporting views shall identify refresh time and source versions.

EDM-000068 - Calculated fields. Calculated values shall preserve formula or rule version and input lineage.

EDM-000069 - Confidence metadata. Estimated or inferred values shall support confidence and review status.

EDM-000070 - Metadata extensibility. Entities shall support governed tags, notes, attachments, and custom fields.

EDM-000071 - Localization. Names, addresses, dates, numbers, units, and currencies shall support locale-aware presentation.

EDM-000072 - Data portability. Core tenant data shall be exportable in documented machine-readable formats.

EDM-000073 - Seed data. Required reference sets shall be supplied as versioned seed data.

EDM-000074 - Test data. Non-production test data shall be synthetic or de-identified.

EDM-000075 - Migration reconciliation. Migrations shall reconcile counts, totals, rejected records, and lineage.

Canonical Entity Catalog

ID	Entity	Domain	Purpose	Lifecycle
ENT-P03-001	Tenant	Foundation	Top-level data isolation and subscription boundary	active; inactive; archived
ENT-P03-002	Organization	Foundation	Operating organization within a tenant	active; inactive; archived
ENT-P03-003	LegalEntity	Foundation	Registered company or legal business entity	active; inactive; archived
ENT-P03-004	BusinessUnit	Foundation	Operational division, branch, or department	active; inactive; archived
ENT-P03-005	Location	Foundation	Reusable physical or virtual location	active; inactive; archived
ENT-P03-006	Project	Work & Delivery	Primary unit of planned and executed work	active; inactive; archived
ENT-P03-007	UserAccount	Foundation	Interactive platform identity	active; inactive; archived
ENT-P03-008	Person	Party & Relationship	Natural person record	active; inactive; archived
ENT-P03-009	Team	Foundation	Named working group	active; inactive; archived
ENT-P03-010	RoleDefinition	Foundation	Governed responsibility and permission role	active; inactive; archived
ENT-P03-011	Party	Party & Relationship	Common customer, vendor, employee, or organization identity	active; inactive; archived
ENT-P03-012	OrganizationParty	Party & Relationship	Organization represented as a party	active; inactive; archived
ENT-P03-013	ContactPoint	Party & Relationship	Email, phone, website, or messaging contact	active; inactive; archived
ENT-P03-014	Address	Party & Relationship	Postal and physical address	active; inactive; archived
ENT-P03-015	PartyRelationship	Party & Relationship	Typed relationship between parties	active; inactive; archived
ENT-P03-016	CustomerAccount	Party & Relationship	Commercial customer account	active; inactive; archived
ENT-P03-017	VendorAccount	Party & Relationship	Supplier or subcontractor account	active; inactive; archived
ENT-P03-018	EmployeeProfile	Party & Relationship	Employment profile linked to a person	active; inactive; archived
ENT-P03-019	Qualification	Safety & Quality	Certification, license, training, or competency	active; inactive; archived
ENT-P03-020	Opportunity	Work & Delivery	Potential work or revenue pursuit	active; inactive; archived
ENT-P03-021	Estimate	Work & Delivery	Versioned cost and price model	active; inactive; archived
ENT-P03-022	EstimateVersion	Work & Delivery	Immutable estimate revision	created; immutable; retained
ENT-P03-023	EstimateLine	Work & Delivery	Detailed quantity, resource, cost, and price line	active; inactive; archived
ENT-P03-024	Proposal	Work & Delivery	Customer-facing offer	active; inactive; archived
ENT-P03-025	Contract	Work & Delivery	Executed agreement governing work	active; inactive; archived
ENT-P03-026	WorkOrder	Work & Delivery	Authorized work package	active; inactive; archived
ENT-P03-027	Task	Work & Delivery	Assignable unit of work	active; inactive; archived
ENT-P03-028	ScheduleActivity	Work & Delivery	Planned activity with dates, logic, and progress	active; inactive; archived
ENT-P03-029	DailyReport	Work & Delivery	Daily field production and condition record	active; inactive; archived
ENT-P03-030	ApprovalRequest	Work & Delivery	Governed approval case	active; inactive; archived
ENT-P03-031	ApprovalDecision	Work & Delivery	Recorded approval outcome	active; inactive; archived
ENT-P03-032	LaborResource	Resources	Labor classification or person capacity	active; inactive; archived
ENT-P03-033	Crew	Resources	Reusable group of labor and equipment	active; inactive; archived
ENT-P03-034	CrewMember	Resources	Crew membership with role and effective dates	active; inactive; archived
ENT-P03-035	EquipmentAsset	Resources	Owned, leased, or rented equipment	active; inactive; archived
ENT-P03-036	FleetVehicle	Resources	Road-licensed fleet vehicle	active; inactive; archived

ID	Entity	Domain	Purpose	Lifecycle
ENT-P03-037	MaterialItem	Resources	Material catalog item	active; inactive; archived
ENT-P03-038	ServiceItem	Resources	Service catalog item	active; inactive; archived
ENT-P03-039	ResourceRate	Resources	Effective labor, equipment, material, or service rate	active; inactive; archived
ENT-P03-040	VendorQuote	Resources	Vendor-specific quote and terms	active; inactive; archived
ENT-P03-041	CostCode	Financial	Standard cost classification	active; inactive; archived
ENT-P03-042	Budget	Financial	Approved financial plan	active; inactive; archived
ENT-P03-043	BudgetLine	Financial	Budget amount by cost, period, or work scope	active; inactive; archived
ENT-P03-044	Commitment	Financial	Purchase order, subcontract, or committed cost	active; inactive; archived
ENT-P03-045	Invoice	Financial	Receivable or payable invoice	active; inactive; archived
ENT-P03-046	Payment	Financial	Money received or paid	active; inactive; archived
ENT-P03-047	CostTransaction	Financial	Actual, accrued, committed, or forecast cost	active; inactive; archived
ENT-P03-048	TaxDefinition	Financial	Tax rule and jurisdiction	active; inactive; archived
ENT-P03-049	CurrencyRate	Financial	Versioned exchange rate	active; inactive; archived
ENT-P03-050	Document	Documents & Records	Logical document record	active; inactive; archived
ENT-P03-051	DocumentVersion	Documents & Records	Immutable document content version	created; immutable; retained
ENT-P03-052	AttachmentLink	Documents & Records	Link between a document and business object	active; inactive; archived
ENT-P03-053	RecordClassification	Documents & Records	Security, retention, and handling classification	active; inactive; archived
ENT-P03-054	ElectronicSignature	Documents & Records	Signature evidence and status	active; inactive; archived
ENT-P03-055	Site	Location & Spatial	Project or customer site	active; inactive; archived
ENT-P03-056	Parcel	Location & Spatial	Property or land parcel reference	active; inactive; archived
ENT-P03-057	SpatialReference	Location & Spatial	Coordinate reference system	active; inactive; archived
ENT-P03-058	GeometryReference	Location & Spatial	Stored or linked geometry representation	active; inactive; archived
ENT-P03-059	StationRange	Location & Spatial	Linear stationing interval	active; inactive; archived
ENT-P03-060	Jurisdiction	Location & Spatial	Governing geographic authority	active; inactive; archived
ENT-P03-061	SafetyIncident	Safety & Quality	Safety or environmental incident	active; inactive; archived
ENT-P03-062	Inspection	Safety & Quality	Structured inspection event	active; inactive; archived
ENT-P03-063	Observation	Safety & Quality	Safety, quality, or field observation	active; inactive; archived
ENT-P03-064	CorrectiveAction	Safety & Quality	Action to resolve a finding	active; inactive; archived
ENT-P03-065	Permit	Safety & Quality	Permit, authorization, or compliance record	active; inactive; archived
ENT-P03-066	AIAgent	AI & Automation	Governed AI agent definition	active; inactive; archived
ENT-P03-067	AIExecution	AI & Automation	Single AI execution record	active; inactive; archived
ENT-P03-068	AIRecommendation	AI & Automation	Structured AI recommendation	active; inactive; archived
ENT-P03-069	AIWriteback	AI & Automation	Proposed or completed AI data change	active; inactive; archived
ENT-P03-070	FeedbackRecord	AI & Automation	Human or outcome feedback on AI/system output	active; inactive; archived
ENT-P03-071	ExternalIdentifier	Integration & Audit	Identifier from another system	active; inactive; archived
ENT-P03-072	SourceSystem	Integration & Audit	External or legacy system definition	active; inactive; archived
ENT-P03-073	DataMapping	Integration & Audit	Field and value mapping specification	active; inactive; archived
ENT-P03-074	ImportBatch	Integration & Audit	Governed data import	active; inactive; archived

ID	Entity	Domain	Purpose	Lifecycle
ENT-P03-075	DomainEvent	Integration & Audit	Versioned business or platform event	created; immutable; retained
ENT-P03-076	DataLineage	Integration & Audit	Origin and transformation lineage	active; inactive; archived
ENT-P03-077	ChangeRecord	Integration & Audit	Versioned record change metadata	active; inactive; archived
ENT-P03-078	AuditEvent	Integration & Audit	Immutable actor-action-object audit event	created; immutable; retained
ENT-P03-079	ReferenceSet	Foundation	Governed code and lookup collection	active; inactive; archived
ENT-P03-080	ReferenceValue	Foundation	Effective-dated lookup value	active; inactive; archived
ENT-P03-081	CustomFieldDefinition	Foundation	Tenant-governed extensible field	active; inactive; archived
ENT-P03-082	CustomFieldValue	Foundation	Typed custom field value	active; inactive; archived

Rules, Workflows, and Engines

- GES-P03-BR-000001 - Tenant boundary.** A record may reference only records belonging to the same tenant unless an approved shared-master pattern applies.
- GES-P03-BR-000002 - Immutable primary key.** Primary identifiers never change after record creation.
- GES-P03-BR-000003 - Version check.** Updates require the current record version.
- GES-P03-BR-000004 - Released version immutability.** Released estimates, contracts, document versions, events, and audit events cannot be edited.
- GES-P03-BR-000005 - Effective date integrity.** Effective start must precede effective end and overlapping active records are prohibited where uniqueness applies.
- GES-P03-BR-000006 - Money context.** A monetary amount must include currency.
- GES-P03-BR-000007 - Quantity context.** A quantity must include compatible unit and dimension.
- GES-P03-BR-000008 - Source preservation.** Imported values preserve source system, external identifier, import batch, and original value where transformed.
- GES-P03-BR-000009 - AI writeback approval.** AI recommendations do not change authoritative records until the required approval is completed.
- GES-P03-BR-000010 - Document checksum.** Accepted document content must have a validated checksum.
- GES-P03-BR-000011 - Party master merge.** A party merge preserves all source identifiers, relationships, addresses, contacts, and reversal evidence.
- GES-P03-BR-000012 - No silent data correction.** Failed quality or validation rules create explicit exceptions or rejected rows.
- GES-P03-BR-000013 - Lifecycle integrity.** Inactive or archived master data cannot be assigned to new work unless policy permits.
- GES-P03-BR-000014 - Audit completeness.** Material create, update, archive, merge, approve, import, and AI writeback actions create audit events.
- GES-P03-BR-000015 - Retention inheritance.** A record inherits retention from classification, entity type, and governing jurisdiction.
- WF-P03-001 - Master Record Creation.** Capture; validate; duplicate check; steward review; assign identifiers; activate; audit
- WF-P03-002 - Master Record Merge.** Select survivor; preserve aliases; move relationships; write lineage; audit; notify
- WF-P03-003 - Schema Change.** Assess compatibility; version schema; migrate; reconcile; certify; publish
- WF-P03-004 - Data Import.** Stage; map; validate; deduplicate; preview; approve; commit; reconcile; archive evidence
- WF-P03-005 - Data Quality Exception.** Create exception; assign; analyze lineage; correct or waive; revalidate; record outcome
- WF-P03-006 - Record Archive and Restore.** Check retention, hold, dependencies; approve; archive or restore; audit
- WF-P03-007 - AI Recommendation Writeback.** Attach evidence; compare current version; authorize; approve; validate; commit; audit; feedback
- WF-P03-008 - Data Export.** Authorize scope; extract canonical data; include lineage; checksum; package; expire
- ENG-P03-001 - Identifier Engine.** Identifiers are stable and non-guessable
- ENG-P03-002 - Reference Resolution Engine.** Tenant, lifecycle, and effective dates must be compatible
- ENG-P03-003 - Master Data Matching Engine.** No automatic merge above configured risk threshold
- ENG-P03-004 - Effective Dating Engine.** Enforce non-overlap where declared
- ENG-P03-005 - Data Quality Engine.** Never silently change source values
- ENG-P03-006 - Lineage Engine.** Every derived value is traceable to inputs and logic
- ENG-P03-007 - Schema Compatibility Engine.** Published contracts require declared compatibility
- ENG-P03-008 - Retention Resolution Engine.** Legal hold overrides expiration
- ENG-P03-009 - Unit and Currency Context Engine.** No implicit cross-context arithmetic
- ENG-P03-010 - AI Writeback Guard Engine.** AI cannot exceed caller or agent authority